

SPEI3 Forecasting Scheme — v2

Anas Soubaha

2026-05-24

SPEI3 Forecasting Scheme — v2

- 1. Inputs (Before Preprocessing)
- 2. Preprocessing
- 3. Feature Engineering
- 4. Cross-Validation Strategy
- 5. Standardization
- 6. Feature Selection (per fold)
- 7. Model Configurations
- 8. Hyperparameter Tuning
- 9. Forecast Generation
- 10. Evaluation Metrics
- 11. Outputs

Summary of Experimental Structure (v2)

SPEI3 Forecasting Scheme — v2

1. Inputs (Before Preprocessing)

Input group	Variable	Temporal coverage	Spatial resolution	Storage	Source
Climatological	Precipitation (mm/mon)	1901–2024	$0.5^{\circ} \times 0.5^{\circ}$	NetCDF	CRU
	Min, Max temperature ($^{\circ}\text{C}$)	1901–2024	$0.5^{\circ} \times 0.5^{\circ}$	NetCDF	CRU
	PET (mm/day)	1901–2024	$0.5^{\circ} \times 0.5^{\circ}$	NetCDF	CRU
	Solar radiation ($\text{J}\cdot\text{m}^{-2}$ / $\text{W}\cdot\text{m}^{-2}$)	1940–2025	$0.25^{\circ} \times 0.25^{\circ}$	NetCDF	ERA5
	Wind speed at 2 m ($\text{m}\cdot\text{s}^{-1}$)	1940–2025	$0.25^{\circ} \times 0.25^{\circ}$	NetCDF	ERA5

Input group	Variable	Temporal coverage	Spatial resolution	Storage	Source
Large-scale climate indices	VPD (hPa)	1940–2025	$0.25^\circ \times 0.25^\circ$	NetCDF	ERA5
	TCWV (kg/m ²)	1940–2025	$0.25^\circ \times 0.25^\circ$	NetCDF	ERA5
	Rootzone Soil Moisture (RZSM, 0–100 cm, m³/m³)	1950–2024	$0.25^\circ \times 0.25^\circ$	NetCDF	ERA5-Land (derived from swvl1–swvl3)
	NAO Index	1950–2026	—	Text	NOAA CPC
	ENSO Niño 3.4	1870–2025	—	NetCDF	NOAA PSL
	Mediterranean Oscillation (MO)	1940–2025	—	NetCDF	ERA5-derived (Gibraltar – Lod SLP, standardized 1981–2010)

Potential additions deferred to a future iteration if v1 underperforms:

- AMO (Atlantic Multidecadal Oscillation) — documented linkage to North African precipitation
- AO (Arctic Oscillation) — partial NAO redundancy, possible additive skill

2. Preprocessing

a. Align time period **1950–2024** for all input variables.

b. Upscale ERA5 $\rightarrow 0.5^\circ$ via conservative area-weighted averaging. All gridded data: 64×64 at 0.5° .

c. Missing-value handling (explicit):

- Temporal interpolation up to 2 consecutive missing months
- Drop cells with $> 5\%$ missing in the training period of any fold

d. RZSM derivation (replaces SSMI in v2):

- Source: ERA5-Land volumetric soil water layers 1–3 (swvl1: 0–7 cm; swvl2: 7–28 cm; swvl3: 28–100 cm)
- Depth-weighted average to 0–100 cm: $RZSM = (7 \cdot swvl1 + 21 \cdot swvl2 + 72 \cdot swvl3) / 100$
- Upscale to 0.5° via conservative area-weighted average (as for the other ERA5 variables)

- **Rationale:** SSMI is pre-standardized, which would introduce subtle look-ahead leakage similar to SPEI3. Using raw RZSM and standardizing fold-wise avoids this.

e. Document SPEI3 computation method in study metadata: distribution (log-logistic), parameter estimator (L-moments), and fitting period. v1 default: full-record fit (1950–2024) with a sensitivity test on the last fold.

f. Stationarity diagnostics (run once, report in supplementary):

- Mann-Kendall trend test on annual mean SPEI3, per cell
- KS test on SPEI3 distributions for 1950–1989 vs 1990–2024
- Diagnostic only — does not gate modeling

3. Feature Engineering

3.1 Target Construction (Lead-Specific)

$y_t = \text{SPEI3}(t + L)$, $L \in \{1, 3, 6\}$ months

Three independent datasets: Dataset_L1, Dataset_L3, Dataset_L6.

- Training: all months 1950–2024
- Evaluation: target month $\in$ [Nov–Feb] only

3.2 Predictor Set at Forecast Issue Time t

(A) Contemporary predictors (time t)

Precip, Tmax, Tmin, PET, VPD, Wind, Solar, TCWV, RZSM, ENSO, NAO, MO, SPEI3

(B) Lag selection methodology

- **PACF as primary tool** (not ACF), computed per CV fold on training data only
- Test lags 1–12 uniformly for **all variables** — let the data prune
- Significance threshold: $|\text{PACF}| > 0.20$, with a sensitivity test at 0.10 and 0.30 reported in appendix
- **CCF screening** between SPEI3 and each candidate predictor on the training fold; include the strongest-lag CCF result if not already selected by PACF
- For linear models: hand all surviving lags to **Lasso / Elastic Net** for final pruning
- For RF / XGBoost: use the PACF + CCF output directly (trees handle redundancy)

Variables explicitly tested for long-memory lags (up to 12): SPEI3, RZSM, ENSO, NAO, MO, TCWV, Precip, VPD. Fast-response variables tested for lag 1 only by default: Tmin, Tmax, Solar, Wind (extendable if PACF justifies).

(C) Seasonal encoding

- $\sin(2\pi \cdot m / 12)$, $\cos(2\pi \cdot m / 12)$ where m is month index

(D) Spatial encoding

- latitude, longitude as features for the **global model** (Section 7.2)

4. Cross-Validation Strategy

5-fold rolling-origin, expanding training window

Fold	Train	Validate	Test
1	1950–1989 (40 y)	1990–1996 (7 y)	1997–2001 (5 y)
2	1950–1996 (47 y)	1997–2001 (5 y)	2002–2006 (5 y)
3	1950–2006 (57 y)	2007–2011 (5 y)	2012–2016 (5 y)
4	1950–2011 (62 y)	2012–2016 (5 y)	2017–2020 (4 y)
5	1950–2016 (67 y)	2017–2020 (4 y)	2021–2024 (4 y)

Leakage controls:

- Target shifting performed before splitting; no future predictors
- Per-fold standardization (Section 5)
- Per-fold feature selection (Section 6)
- Per-fold PACF / CCF lag selection (Section 3.2 B)

Aggregation across folds:

- **Pooled:** concatenate predictions across folds → compute headline metric once
- **Per-fold:** same metrics reported per fold in supplementary table to show stability

5. Standardization

Fold-wise standardization using **training-period statistics only**.

Pre-standardized exception list — do NOT re-standardize:

- ENSO, NAO, MO (and their lags)
- **SPEI3 (and its lags)** when used as a predictor

All other variables — **including RZSM** — are standardized fold-wise.

Procedure per fold:

1. Compute mean / std on training data only
2. Apply to training, validation, and test sets

6. Feature Selection (per fold)

Method	Applied to	Purpose
VIF (threshold < 5)	Linear-family pre-filter	Remove multicollinearity
Lasso / Elastic Net	Final feature set for linear models	L1 sparsity, lag pruning
Permutation importance	RF	Rank features, report top- k
SHAP values	XGBoost	Interpretability and ranking
RFE	Tabular models (optional)	Subset-selection sensitivity

Pipeline per model family:

- Linear / Ridge / Lasso / Elastic Net: VIF filter → Lasso / Elastic Net during fit
- RF: all features in; permutation importance for diagnostics
- XGBoost: all features in; SHAP for diagnostics

7. Model Configurations

7.0 Baselines

Baseline	Forecast formula	Purpose
Climatology	per-cell per-calendar-month training mean	Reference for skill scores
Persistence	$\text{SPEI3}(t+L) = \text{SPEI3}(t)$	Inertia benchmark
Damped persistence	$\text{SPEI3}(t+L) = \alpha^L \cdot \text{SPEI3}(t)$, $\alpha = \text{lag-}L$ autocorrelation	Optimal AR(1), strong mid-lead benchmark
AR(p)	Linear regression on lagged SPEI3 only	Univariate AR benchmark
Teleconnection-only (optional)	Linear regression on ENSO / NAO / MO lags only	Skill from large-scale drivers alone

7.1 ML Models

Model	Input	Purpose
Linear (OLS)	Tabular	Statistical baseline
Ridge	Tabular	L2-regularized linear
Lasso	Tabular	L1-regularized (also feature selector)
Elastic Net	Tabular	L1 + L2 regularized

Model	Input	Purpose
Random Forest	Tabular	Nonlinear ensemble
XGBoost	Tabular	Gradient boosting (primary tabular benchmark)
LSTM	Lagged sequences	Temporal memory (deferred)
CNN	Spatial grids	Spatial pattern extraction (deferred)
CNN-LSTM / ConvLSTM	Spatio-temporal sequences	Joint spatio-temporal (deferred)

7.2 Modeling Unit

- **Default:** global model with (lat, lon) as features — one model per (model family, fold), trained on all 64×64 cells jointly
- **Sensitivity test:** per-cell models for the best-performing family

8. Hyperparameter Tuning

Protocol A — tune on validation, refit on train + val, evaluate on test:

```
for fold in cv_folds:
    best_hp = search(model_type, X_train, y_train, X_val, y_val)
    final_model = model_type(best_hp).fit(X_train u X_val, y_train
u y_val)
    preds = final_model.predict(X_test)
```

Search method:

- **Grid search:** Ridge, Lasso, Elastic Net
- **Bayesian (Optuna):** RF, XGBoost — 50–100 trials each

Suggested search spaces:

Model	Space
Ridge / Lasso	$\alpha \in \text{logspace}(-3, 3, 13)$
Elastic Net	$\alpha \in \text{logspace}(-3, 3, 7);$ $\text{l1_ratio} \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$
RF	$\text{n_estimators} \in \{200, 500, 1000\}; \text{max_depth} \in \{\text{None}, 5, 10, 20\}; \text{min_samples_leaf} \in \{1, 5, 20\}; \text{max_features} \in \{\text{sqrt}, 0.5, 1.0\}$
XGBoost	n_estimators via early stopping on val; $\text{max_depth} \in \{3, 6, 10\}; \text{lr} \in \{0.01, 0.05, 0.1\}; \text{subsample}, \text{colsample_bytree} \in \{0.5, 0.8,$

Model	Space
	1.0}; reg_alpha, reg_lambda ∈ logspace(-3, 1, 5); min_child_weight ∈ {1, 5, 20}

9. Forecast Generation

For each issue month t :

1. Extract predictors up to month t
2. Apply trained model for lead L (one model per (family, fold, lead))
3. Produce forecast $\widehat{\text{SPEI3}}(t + L)$
4. Retain forecasts verifying in Nov–Feb only

10. Evaluation Metrics

10.1 Deterministic (continuous SPEI3)

Metric	Formula	Use
RMSE	$\sqrt{[\text{mean}((\hat{y} - y)^2)]}$	Error magnitude
MAE	$\text{mean}(\hat{y} - y)$	
Pearson r	$\text{corr}(\hat{y}, y)$	Pattern correlation
ACC	$\text{corr}(\hat{y} - \text{clim}, y - \text{clim})$	Headline — anomaly correlation
MSSS vs climatology	$1 - \text{MSE}(\text{model}) / \text{MSE}(\text{climatology})$	Headline — % improvement over climatology
MSSS vs persistence	$1 - \text{MSE}(\text{model}) / \text{MSE}(\text{persistence})$	Added value beyond inertia

10.2 Categorical (drought classes — McKee thresholds)

Class	SPEI3
Extreme drought	< -2.0
Severe drought	$[-2.0, -1.5)$
Moderate drought	$[-1.5, -1.0)$
Normal / wet	≥ -1.0

Metric	Use
POD (hit rate)	Drought detection rate
FAR	False alarm ratio
CSI	Combined accuracy
HSS	Headline — binary at $\text{SPEI3} < -1.0$ and multi-class

Metric	Use
ETS	Chance-corrected hits

10.3 Probabilistic (optional)

CRPS, Brier skill score, reliability diagrams — required only if uncertainty quantification is added (recommended via quantile XGBoost in a follow-up).

10.4 Headline Metrics (committed)

1. ACC
2. MSSS vs climatology
3. HSS for SPEI3 < -1.0

10.5 Reporting Structure

- Per lead time ($L = 1, 3, 6$)
- Per CV fold (supplementary table) + pooled across folds (headline)
- Spatial: per-cell skill maps + pooled metrics across cells
- **Block bootstrap 95 % CIs** on all headline metrics:
 - Stationary bootstrap; mean block ≈ 12 months
 - For winter-only metrics: year-blocks (full Nov–Feb season per block)
 - 1000 replicates

11. Outputs

Output	Description
Grid-level forecasts	0.5° SPEI3 predictions per (model, lead)
Spatial skill maps	Per-cell ACC, MSSS, HSS for each (model, lead)
Forecast-vs-truth time series	At selected cells (Casablanca, Marrakech, Agadir)
Feature importance diagnostics	Permutation (RF), SHAP (XGBoost), per lead
Skill comparison tables	Models $\times$ leads $\times$ headline metrics, with bootstrap CIs
Per-fold stability tables	Same metrics broken by fold
Baseline-vs-ML skill plots	Bars showing MSSS for each model relative to each baseline
Lag selection diagnostics	PACF / CCF plots per variable per fold
Stationarity diagnostics	Mann-Kendall and KS test results per cell

Summary of Experimental Structure (v2)

Dimension	Specification
Time period	1950–2024
Temporal resolution	Monthly
Spatial resolution	64 × 64 grid at 0.5° over Morocco
Target	SPEI-3 at L = 1, 3, 6 months
Evaluation season	Winter (Nov–Feb)
Cross-validation	5-fold rolling-origin, expanding train window
Leakage control	Fold-wise standardization, feature selection, lag selection; strict target shifting
Pre-standardized exceptions	ENSO, NAO, MO, SPEI3 (as predictor)
Climate drivers	ENSO, NAO, MO (+ optional AMO, AO in v2)
Lag selection	PACF + CCF, threshold $ \text{PACF} > 0.20$, sensitivity $\in \{0.10, 0.30\}$, Lasso finalizes for linear
Modeling unit	Global model with (lat, lon) features (v1); per-cell as sensitivity
Baselines	Climatology, persistence, damped persistence, AR(p)
ML models	OLS / Ridge / Lasso / Elastic Net, RF, XGBoost
DL models	LSTM, CNN, CNN-LSTM (or ConvLSTM) — deferred
HP tuning	Protocol A — tune on val, refit on train + val, eval on test
HP search	Grid (linear), Optuna (RF, XGBoost); early stopping (XGBoost)
Deterministic metrics	RMSE, MAE, ACC, MSSS-vs-clim, MSSS-vs-persistence
Categorical metrics	POD, FAR, CSI, HSS, ETS — binary at SPEI3 < −1.0 + multi-class
Headline metrics	ACC, MSSS-vs-clim, HSS at SPEI3 < −1.0
Uncertainty	Stationary block bootstrap, 1000 replicates, 95 % CI
Outputs	Forecasts, spatial skill maps, time series, importance, skill tables, CIs